Behavioral Adjustments and Offensive Performance Under High Leverage in Major League Baseball

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I. Introduction

In baseball, not all plate appearances carry equal weight. Leverage Index (LI), created by Tom Tango, measures the importance of a particular plate appearance by quantifying the extent to which win probability could change on that event [1]. Situations with an LI above 2.0 are considered high leverage, representing moments where a single plate appearance can drastically shift a team's probability of winning. These situations typically appear in the later innings of close games. An LI below 1.0 indicates a low leverage situation that has minimal impact on game outcome. These situations typically occur early in games or when there is a large score difference. The prevailing view in baseball analytics holds that clutch performance is largely attributed to random variance rather than a repeatable skill. This project aims to statistically evaluate whether performance differences in high leverage situations are systematic, and whether they are associated with measurable changes in batter behavior.

Data for this analysis was sourced from TruMedia Networks [2], a sports analytics platform widely used across professional baseball organizations. The dataset contains Major League Baseball statistics spanning the 2023 season through the 2025 season, restricted to hitters with sufficient plate appearance to produce stable estimates across both high and low leverage splits (n = 143 players). By the nature of the game, high and low leverage plate appearance accumulate at very different rates. On average, hitters accumulated 298 high leverage plate appearances over the three season window (range: 185-402), compared to an average of 1,376 low leverage plate appearances (range: 1,022-1,795). For each player, performance and plate discipline metrics were collected under both leverage conditions. Offensive performance was measured using expected On-base Plus Slugging (xOPS), which estimates offensive production based on quality of contact rather than outcomes alone. xOPS was selected as the primary offensive performance metric as it captures both on-base ability and power production while reducing the influence of luck. Plate discipline was captured through three behavioral metrics: Chase Rate, the percentage of pitches swung at outside the strike zone; Zone Swing Rate, the percentage of pitches swung at inside the strike zone; Whiff Rate, the percentage of swings that make no contact.

A preliminary examination of the data reveals, as expected, that hitters produce lower xOPS in high leverage situations than in low leverage situations. Across the full sample, hitters produce a mean xOPS of **0.725** in high leverage situations compared to **0.758** in low leverage situations, representing a mean decline of 0.033 xOPS points. While this difference may appear small in absolute terms, a decline of this magnitude is considered practically significant within the context of baseball. This decline is not uniform across players, while the majority of hitters experience a drop in xOPS under high leverage conditions, some players maintain their performance level and others improve relative to their low leverage baseline. Suggesting that individual responses to high pressure situations vary meaningfully across the league. Table I provides a descriptive overview of the mean value for all key metrics across both leverage condition. This variation raises the central question driving this analysis: can performance differences in high leverage situation be explained by measurable changes in batter behavior?

To address this question, three statistical analyses are conducted. First, a paired t-test evaluates whether the league-wide decline in xOPS is statistically significant. Second, a repeated measures ANOVA examines whether the plate discipline metrics shift systematically under pressure at the league level. Finally, a linear regression model tests whether changes in plate discipline predict the degree to which an individual hitter's performance declines relative to the league average, directly evaluating whether behavioral adjustments explain clutch performance differences across players.

II. Paired t-Test Analysis of Offensive Performance Under High and Low Leverage

A. *Test Method*

A paired t-test was conducted to compare player-level offensive performance (xOPS) between high and low leverage situations. This test is appropriate because each player in the dataset is observed under both conditions, creating dependent observations and allowing analysis of within-player differences. The test determines whether there is a statistically significant difference in mean xOPS between high and low leverage situations.

$$H_0 : \mu_{\text{High}} = \mu_{\text{Low}}$$

TABLE I: Descriptive Statistics: High vs. Low Leverage Situations (n = 143 Players)

Metric	High Leverage	Low Leverage	Difference
Volume			
Plate Appearances	298.3	1,375.8	-1,077.5
Plate Discipline (%)			
Zone Swing Rate	68.58	66.36	+2.22
Chase Rate	30.55	28.81	+1.75
Whiff Rate	23.00	21.36	+1.64
Performance			
xOPS	0.725	0.758	-0.033

Note: Difference = High Leverage − Low Leverage. Rate statistics reported as percentages.

$$H_a : \mu_{\text{High}} \neq \mu_{\text{Low}}$$

Where μ represents the mean xOPS for each leverage group across players.

B. Assumptions

The response variable (xOPS) is continuous, satisfying the requirement for applying a paired t-test. Assumptions for the paired t-test were evaluated prior to analysis. Normality of the within-player differences in xOPS was assessed using a normal probability plot (Fig. 1). The points followed an approximately linear pattern, and the corresponding normality test yielded a p-value of 0.327. Therefore, there is no evidence at the 5% significance level to conclude that this is not a normally distributed population. This supports the assumption that the sample of differences can be treated as approximately normally distributed for the purposes of the paired t-test. A box plot of the differences (Fig. 2) identified two outliers (one high and one low); however, sensitivity analysis confirmed that the removal of these points did not affect the results. Independence of observations across players was assumed based on the study design. Each player is treated as a separate observational unit, and while game context may introduce some shared external influences, these effects are not expected to create a systematic dependence between player's performance measurements over a large sample at the league level.

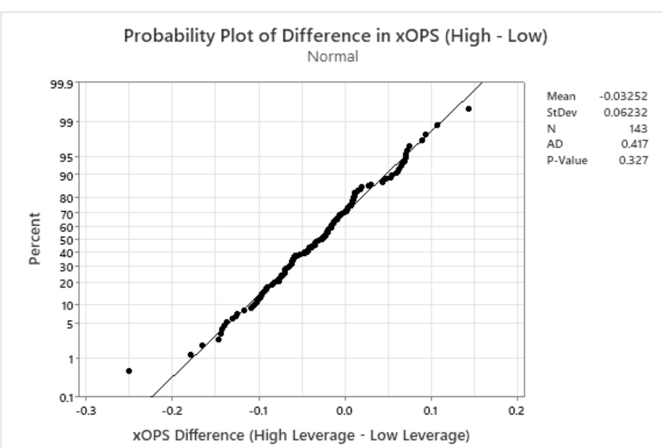


Fig. 1: Normal probability plot of within-player xOPS differences, indicating no substantial deviation from normality.

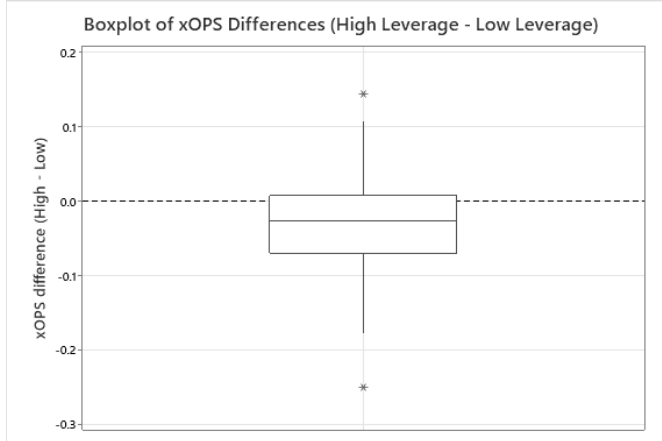


Fig. 2: Box plot of within-player xOPS differences (High − Low leverage). Negative values indicate decreased performance in high leverage situations.

C. Results

The paired t-test revealed a statistically significant difference in xOPS. With a p-value < 0.0001, there is sufficient evidence to reject the null hypothesis and conclude that mean xOPS differs significantly between high and low leverage situations. The mean within-player difference was -0.033, indicating decreased offensive performance in high leverage situations. The 95% confidence for the mean difference was $(-0.04283, -0.02222)$, which does not include zero, further supporting the presence of a significant decrease in performance. The standardized effect size was moderate (Cohen's $d = -0.5218$), indicating a consistent directional shift across players.

D. Discussion

The results indicate a consistent decline in offensive performance under high leverage situations, compared to low leverage baseline performance. The moderate effect size suggests a meaningful shift in player-level outcomes. While the mean change in xOPS is relatively small in absolute terms, the standardized effect indicates that this decrease is consistent across a large proportion of players rather than being driven by a small subset of extreme observations. From a baseball context, high leverage plate appearances often occur against a team's most effective relief pitchers, who are typically used in late-game, high-pressure situations. The observed decline in offensive performance may therefore reflect both situational pressure and increased pitcher quality. This motivates further analysis of whether hitters adjust their approach and plate discipline behavior under high leverage conditions, and whether these adjustments help explain the decline in performance.

III. Repeated Measures ANOVA of Plate Discipline Behavior

A. Test Method

A repeated measure ANOVA was conducted to compare player-level plate discipline behavior between high and low leverage situations. Three plate discipline metrics were evaluated: Zone Swing Rate, Chase Rate, and Whiff Rate. This test is appropriate as each player in the dataset is observed under both conditions, creating dependent observations and allowing analysis of within-player differences in approach. The repeated measures design accounts for the natural correlation between each player's high and low leverage measurements and isolates the effect of leverage on hitter behavior. A separate repeated measure ANOVA was conducted for each of the three selected metrics, allowing independent evaluation of how leverage influences each of these components of hitter approach and behavior. The test determines whether there is a statistically significant difference in the mean rate of each plate discipline metric between high and low leverage situations.

For each metric:

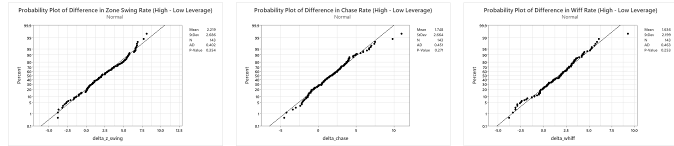
$$H_0 : \mu_{\text{High}} = \mu_{\text{Low}}$$

$$H_a : \mu_{\text{High}} \neq \mu_{\text{Low}}$$

Where μ represents the mean difference rate of the corresponding plate discipline metric (Zone Swing Rate, Chase Rate, or Whiff Rate) across all players.

B. Assumptions

The assumptions for a repeated measures ANOVA were evaluated prior to analysis. The response variables are continuous percentages, satisfying the requirement that the dependent variable be measured on a continuous scale. Normality of the within-player differences was assessed for each of the three metrics using normal probability plots (Fig. 3). The points followed approximately linear patterns across all three plots, and the corresponding normality tests yielded p-values greater than 0.05 in each case. Therefore, there is no evidence at the 5% significance level to conclude that the within-player differences depart from a normal distribution. This supports the assumption that differences for each metric can be treated as approximately normal. Sphericity is automatically satisfied as the leverage factor has only two levels (high and low). Box plots of the within-player differences (Fig. 4) identified two high outliers in Chase Rate differences and one high outlier in Whiff Rate differences. Removing the outliers from the analysis did not change conclusions for either metric, confirming that the results are not driven by these extreme outliers. Given the within-subject design, formal tests of variance equality across metrics were not required. Independence of observations between players was assumed based on the study design, consistent with the rationale outlined in Section II. Within-player observations are intentionally correlated under the repeated measures framework, and this correlation is appropriately modeled by the test rather than treated as a violation of independence.



(a) Zone Swing Rate (b) Chase Rate (c) Whiff Rate

Fig. 3: Normal probability plots of within-player differences for the three plate discipline metrics. All three sets of differences pass the normality assessment at the 5% significance level.

C. Results

The repeated measures ANOVA revealed statistically significant differences in plate discipline behavior between high and low leverage situations for all three metrics. Zone Swing Rate showed the strongest effect (F(1,142) = 10.83, p = 0.001), followed by Whiff Rate (F(1,142) = 6.08, p = 0.014) and Chase Rate (F(1,142) = 5.96, p = 0.015). With p-values below 0.05 in each case, there is sufficient evidence to reject the null hypothesis and conclude that mean Zone Swing Rate, Chase Rate, and Whiff Rate each differ significantly between leverage conditions. The mean within-player differences indicated a consistent directional shift across all three behaviors. Hitters increased their Zone Swing Rate by an average of 2.219 percentage points in high leverage situations (68.578% vs 66.359%), increased their Chase Rate by 1.749 percentage points (30.555% vs 28.806%), and increased their Whiff Rate by 1.636 percentage points (22.997% vs 21.361%). Although each effect was statistically significant, the proportion of variance in plate discipline behavior explained by the leverage condition was small for all three metrics. The R^2 values were 3.67% for Zone Swing Rate, 2.10% for Whiff Rate, and 2.05% for Chase Rate, indicating that leverage condition accounts for only a small portion of the total variation in hitter approach.

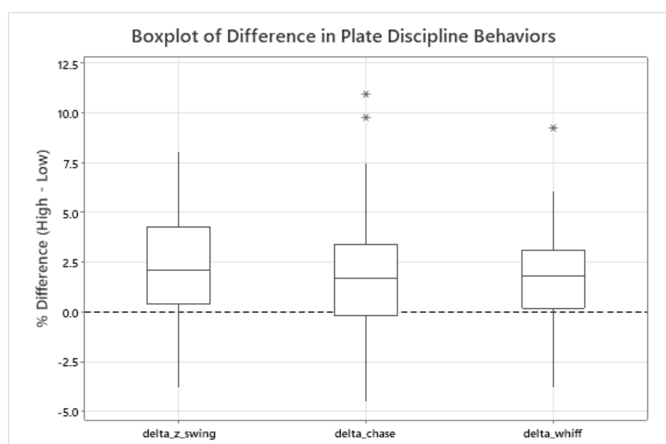


Fig. 4: Box plots of within-player differences (High − Low leverage) for Zone Swing Rate, Chase Rate, and Whiff Rate. Positive values indicate increased rates in high leverage situations.

TABLE II: Repeated Measures ANOVA Results by Plate Discipline Metric

Metric	F(1, 142)	p-value	R^2
Zone Swing Rate	10.83	0.001	3.67%
Chase Rate	5.96	0.015	2.05%
Whiff Rate	6.08	0.014	2.10%

The remaining variation is attributed to factors outside the scope of this test.

D. Discussion

The repeated measures ANOVA results indicate that hitters do change their approach under pressure, but the magnitude of that change is modest. Across all three metrics, the direction of the shift is the same, hitters swing more often and make less contact. In plain terms, hitters become more aggressive but less effective in high leverage situations. Notably, Zone Swing Rate produced the strongest effect, suggesting that the most pronounced behavioral change is an increased willingness to swing at strikes. Hitters may alter their approach as they feel that in high leverage situations they need to put the ball in play and cannot afford to strike out. While each of these effects is statistically significant, the small R^2 values, all under 4%, indicate that leverage condition explains only a small portion of the variation in hitter behavior. This is an important interpretive point: the statistical detectability of an effect does not imply that the effect is the dominant force shaping the metric. The remaining variation reflects context that this test does not incorporate, including pitcher matchup, the count and base-out state of each plate appearance, and the baseline differences in individual hitter approaches. Leverage, in other words, nudges the league-wide behavior in a consistent direction, but the effect operates on top of a much larger source of variability. The combination of statistically significant league-wide shifts with small effect sizes reinforces the idea that aggregate behavioral changes alone are unlikely to fully explain the decline in xOPS observed in Section II. The league-level averages here do not capture individual variation. Understanding whether a player's specific behavioral adjustments translate into better or worse outcomes requires shifting the analysis from the league level to the player level. This motivates the linear regression model in the next section.

IV. Linear Regression Analysis of Behavioral Adjustments of Performance Differential

A. Test Method

A multiple linear regression was conducted to evaluate whether player-level changes in plate discipline predict the magnitude of player-level changes in offensive performance between high and low leverage situations. This analysis shifts the unit of analysis from the league to the individual hitter, complementing the league-wide findings established in Sections II and III. The regression is appropriate because each player contributes a single observation consisting of one response value and three predictor values, allowing direct estimation of how behavioral adjustments relate to performance differential at the player level. The response variable is an adjusted measure of each player's performance change between leverage conditions. Because the league-wide findings from Section II established that majority of hitters experience a decline in xOPS under high leverage, the raw within-player differential is dominated by this shared directional trend and does not directly capture whether a given hitter performed better or worse than the typical hitter. To isolate individual variation around the league pattern, the response variable is defined as each player's xOPS differential adjusted for the median league differential:

$$\Delta\text{xOPS}_i^{adj} = (\text{xOPS}_{\text{High},i} - \text{xOPS}_{\text{Low},i}) - \text{median}(\Delta\text{xOPS}) \tag{1}$$

A value of zero indicated that the player's change in performance under high leverage matches the typical league-wide change. Positive values indicate that a player performed better relative to the typical hitter, and negative values indicate a player performed worse. This adjustment recenters the response on individual deviation from the league pattern rather than absolute performance change. This aligns more closely with the underlying research question of whether clutch performance reflects player-specific skill rather than the shared league-wide effect.

The three predictors are the corresponding within-player changes in plate discipline behavior:

$$\Delta\text{ZoneSwing}_i = \text{ZoneSwing}_{\text{High},i} - \text{ZoneSwing}_{\text{Low},i} \tag{2}$$

$$\Delta\text{Chase}_i = \text{Chase}_{\text{High},i} - \text{Chase}_{\text{Low},i} \tag{3}$$

$$\Delta\text{Whiff}_i = \text{Whiff}_{\text{High},i} - \text{Whiff}_{\text{Low},i} \tag{4}$$

The regression model takes this form:

$$\Delta\text{xOPS}_a dj_i = \beta_0 + \beta_1 \Delta\text{ZoneSwing}_i + \beta_2 \Delta\text{Chase}_i + \beta_3 \Delta\text{Whiff}_i + \varepsilon_i \tag{5}$$

Where ϵ_i represents the residual error for player i. The intercept $\beta 0$ estimates the expected change in xOPS for a hitter whose plate discipline metrics remain unchanged across leverage conditions, and each slope coefficient estimates the expected change in xOPS associated with a one-percentage-point increase in the corresponding metric, holding the other predictors constant.

Two sets of hypotheses are evaluated. The overall model significance is tested using an F-test:

$$H_0 : \beta_1 = \beta_2 = \beta_3 = 0$$

$$H_a : \text{at least one } \beta_j \neq 0$$

The individual predictor contributions are tested using t-tests on each coefficient:

$$H_0 : \beta_j = 0$$

$$H_a : \beta_j \neq 0$$

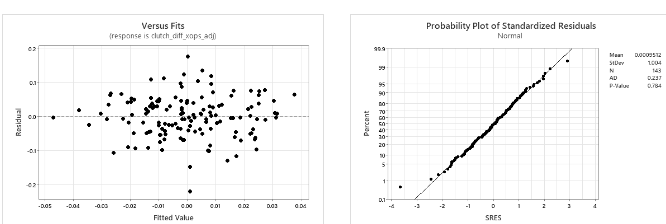
for j = 1,2,3. Together, these tests determine whether behavioral adjustments collectively explain variation in performance differential and whether each individual adjustment carries independent predictive value.

B. Assumptions

The assumptions for multiple linear regression were evaluated prior to analysis. The response variable (adjusted ΔxOPS) and all three predictor variables (ΔZoneSwing, ΔChase, ΔWhiff) are continuous, satisfying the requirement that both the response and predictors be measured on a continuous scale. Linearity was assessed using a residual vs. fitted values plot (Fig. 5a). The plot showed a roughly diamond-shaped pattern with residuals scattered around zero and no systematic trend, funnel, or curvature. This supports both the assumption of a linear relationship between the predictors and the response and the assumption of constant residual variance across the range of fitted values. Normality of the residuals was assessed using a normal probability plot (Fig 5b). The points followed an approximately linear pattern, and the corresponding normality test yielded a p-value of 0.784. Therefore, there is no evidence at the 5% significance level to conclude that the residuals depart from a normal distribution. Multicollinearity was assessed using Variance Inflation Factor (VIF) values for each predictor. The VIF values ranged from 1.03 to 1.11, well below the conventional threshold of 5 used to indicate problematic multicollinearity. This indicates that the three plate discipline change variables provide largely independent information. Independence of observations across players was assumed based on the study design, consistent with the rationale outline in Section II.

C. Results

The multiple linear regression produced a statistically significant overall model (F(3,139) = 3.38, p = 0.020), providing sufficient evidence to reject the null hypothesis and conclude that the set of plate discipline changes carries some predictive value for adjusted xOPS differential. However, the proportion of variance explained by the model was small (R^2 = 6.81%), indicating that the behavioral changes account for only a small portion of the variation in individual performance differential. The individual coefficient tests produced mixed results. ΔWhiff was the only predictor to reach statistical significance at the 5% level, returning a coefficient of $\beta 3$ = -0.0058 (SE = 0.0024, t = -2.15, p = 0.033). The negative sign indicates that with each one-percentage-point increase in ΔWhiff associated with a 0.0058-point decrease in adjusted xOPS differential, holding the other predictors constant. ΔChase returned $\beta 2$ = -0.0038 (SE = 0.0020, t = -1.86, p = 0.065), narrowly missing significance at the 5% level but pointing in the same direction ΔWhiff: hitters who chased more under high leverage tended to underperform relative to the league. ΔZoneSwing returned a coefficient of $\beta 1$ = 0.0021 (SE = 0.0020, t = 1.09, p = 0.279), showing no detectable relationship with performance differential.



(a) Residuals vs. Fitted (b) Normal Probability Plot

Fig. 5: Regression diagnostic plots: (a) residuals vs. fitted values showing no systematic pattern or funnel shape, supporting linearity and constant variance, and (b) normal probability plot of residuals showing approximately linear behavior, supporting normality.

D. Discussion

The regression results indicate that behavioral changes have a measurable but limited relationship with individual performance differential under high leverage. The model achieved overall statistical significance, and ΔWhiff emerged as a significant individual predictor with ΔChase narrowly missing the conventional 5% threshold. Both coefficients are negative, indicating that hitters who became more prone to swing-and-miss under pressure tended to underperform the league median change in xOPS. This pattern is consistent with baseball intuition: chasing pitches outside of the strike zone and missing pitches a hitter does swing at are direct contributors to negative offensive outcomes. At the same time, the small R^2 of 6.81% indicates that even these meaningful predictors capture only a small fraction of the variation in individual performance. The negligible multicollinearity confirms that the limited explanatory power is not a statistical artifact but reflects genuinely weak associations between linear behavior change and individual deviation from the league trend. There is a large number of factors not included in the model that likely contribute to the unexplained variation. The mixed pattern of results, a significant overall model and one significant predictor paired with a small effect size, suggests that the relationship between behavior and performance changes may not be uniform across the player population. It is possible that behavioral changes operate differently in hitters who succeed under pressure than in those who do not. This motivates the final analysis, which groups players based on whether they outperformed, or underperformed the league median change in xOPS, and compared behavioral profiles across these groups using two-sample t-tests. This subgroup approach allows for the possibility that the behavioral signatures of clutch and non-clutch performers differ in ways that aggregate regression coefficients do not reveal.

V. Two-Sample t-Test Analysis of Behavioral Changes by Performance Group

A. Test Method

A series of two-sample t-tests was conducted to compare plate discipline behavioral changes between hitters who outperformed and hitters who underperformed the league median change in xOPS under high leverage. Players were grouped into two categories: Outperformers, those with adjusted ΔxOPS above zero, and Underperformers, those with adjusted ΔxOPS below zero. This analysis tests whether the magnitude or direction of behavioral adjustments differs systematically between players who handle high leverage situations relatively well and players who do not, allowing the relationship between behavior and performance to be examined through a subgroup lens that the linear regression in Section IV cannot fully capture. A separate two-sample t-test was conducted for each of the three plate discipline change variables. For each metric:

$$H_0 : \mu_{\text{Outperform}} = \mu_{\text{Underperform}}$$

$$H_a : \mu_{\text{Outperform}} \neq \mu_{\text{Underperform}}$$

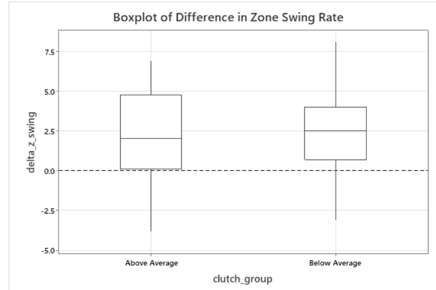
Where μ represents the mean within-player change in the corresponding plate discipline metric for each performance group.

B. Assumptions

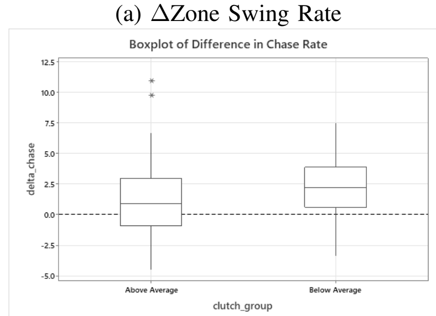
The assumptions for the two-sample t-test were evaluated prior to analysis. The response variables (ΔZoneSwing, ΔChase, and ΔWhiff) are continuous, satisfying the requirement that the response be measured on a continuous scale. Normality within each group was assessed using Anderson-Darling normality tests on each behavioral change variable separately for Outperformers and Underperformers. All six tests returned p-values above 0.05, supporting the assumption that the within-group distributions can be treated as approximately normal for the purposes of inference. Equal variances between groups were assessed using Bonett and Levene tests of variance equality for each behavioral metric. The variance test for ΔZoneSwing returned a p-value of 0.044, indicating unequal variances between groups; consequently, Welch's t-test was applied to that metric. The variance tests for ΔChase (p = 0.501) and ΔWhiff (p = 0.771) returned p-values well above 0.05, supporting the equal-variance assumption, and the pooled-variance two-sample t-test was applied to those metrics. Independence of observations between groups is satisfied by construction, as each player is assigned to exactly one performance group based on their adjusted ΔxOPS value. Independence within each group follows from the same study design rationale used in Sections II through IV.

C. Results

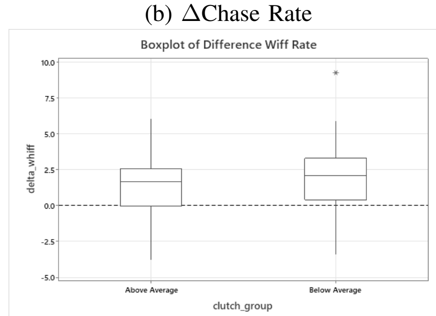
The two-sample t-tests revealed statistically significant differences between Outperformers and Underperformers on two of the three plate discipline change variables. ΔChase produced the strongest effect ($t(141) = -2.33$, $p = 0.021$), followed by ΔWhiff ($t(141) = -2.29$, $p = 0.024$). Both p-values fall below the 0.05 significance threshold, providing sufficient evidence to reject the null hypothesis and conclude that mean behavioral changes differ between performance groups for each of these metrics. The negative t-statistics indicate that Outperformers had smaller (or more negative) increases in chase and whiff rates under high leverage than Underperformers did. ΔZoneSwing did not produce a significant difference between groups ($t(136) = -0.71$, $p = 0.481$), indicating that changes in zone swing rate are not associated with whether a hitter outperforms or underperforms the league median change in xOPS. This result is consistent with the regression finding in Section IV, where ΔZoneSwing also failed to reach significance as a predictor of performance differential. Mean ΔChase was 1.24% in Outperformers compared to 2.26% in Underperformers, a difference of -1.02 percentage points (95% CI: $[-1.887, -0.152]$). Mean ΔWhiff was 1.23% in Outperformers compared to 2.05% in Underperformers, a difference of -0.82 percentage points (95% CI: $[-1.545, -0.112]$).



(a) ΔZone Swing Rate



(b) ΔChase Rate



(c) ΔWhiff Rate

Fig. 6: Box plots of within-player changes in plate discipline metrics by performance group (Outperform vs. Underperform). Outperformers exhibit smaller increases in Chase Rate and Whiff Rate than Underperformers,while Zone Swing Rate changes do not differ meaningfully between groups.

D. Discussion

The two-sample t-test results provide the clearest behavioral signal observed in this study. Hitters who outperformed the league median change in xOPS under high leverage exhibited smaller increases in both chase rate and whiff rate than hitters who underperformed. This pattern indicates that the behavioral signature of relative success under pressure is not avoiding aggressiveness entirely (zone swing rate did not differ between groups) but rather maintaining contact and pitch selection discipline as leverage increases. Outperformers swung at non-strikes less often and missed less often when they did swing, even as their overall swing rate inside the zone shifted at a level comparable to Underperformers. This finding extends and clarifies the linear regression result from Section IV. The regression detected ΔWhiff as a significant predictor of adjusted xOPS differential and indicated that ΔChase narrowly missed conventional significance at the 5% level. The subgroup analysis confirms both relationships with stronger statistical evidence by reformulating the question as a comparison between meaningful player categories rather than as a continuous prediction across the full sample. The directional consistency across both analyses, with whiff and chase increases associated with poorer relative performance and zone swing change carrying no detectable association, strengthens the inference that contact-related behavioral adjustments matter more than overall swing aggressiveness in shaping individual outcomes under high leverage. Taken together with the earlier analyses, the results suggest a coherent picture of clutch performance at the league level. Hitters as a group decline in performance under high leverage (Section II), behave more aggressively (Section III), and the magnitude of those behavioral changes carries some predictive value for individual performance differential (Section IV). The subgroup analysis presented here identifies the specific behaviors that distinguish hitters who handle high leverage relatively well from those who do not: smaller increases in chase rate and whiff rate, rather than smaller increases in zone swing rate. While this study cannot determine whether these behavioral patterns are causes or consequences of relative success, they provide a measurable behavioral profile of hitters who outperform expectations under pressure.

VI. Conclusion

This study evaluated whether differences in offensive performance in high leverage situations are systematic and whether they are associated with measurable changes in batter behavior. The combined results from four analyses converge on a single, somewhat counterintuitive answer: clutch performance, to the extent that it is detectable in plate discipline data, is defined less by what successful hitters do under pressure and more by what they refrain from doing. The paired t-test establishes that batters as a whole experience a small but statistically significant decline in xOPS under high leverage, with moderate effect size (Cohen's $d = -0.5218$). The repeated measures ANOVA confirmed that hitters systematically shift their approach in these situations, swinging more often both inside and outside of the strike zone and making contact less often. The linear regression produced a significant overall model with limited explanatory power (R^2 = 6.81%). The two-sample t-tests sharpened the finding by demonstrating that hitters who outperformed the league median exhibited significantly smaller increases in chase and whiff rates that hitters who underperformed, while changes in zone swing rate did not differ between groups.

The behavioral signature of relative success under pressure is not a positive adjustment but the absence of one. Outperformers are essentially the same hitters in high leverage that they are in low leverage. Their chase and whiff rates rise less, and their overall approach drifts less from the baseline they have already proven works for them. Underperformers, by contrast, exhibit the league-wide behavioral patterns in exaggerated form. They expand the zone more, swing and miss more often, and let the situation pull them away from their established approach. In this framing, "clutch" is less about rising to the occasion and more about staying resisting the gravitational pull of the moment away from who you are. The real skill being measures is behavioral stability under pressure, the ability to keep doing what was already working when the stakes raise. This reframing has practical implications. It suggests that clutch performance should not be treated as an innate quality but as a measurable, and potentially trainable, form of plate discipline consistency. A hitter's high leverage profile cannot be evaluated against an aspirational standard of exceptional performance, but against their own low leverage baseline, with deviation from baseline serving as the diagnostic signal. The behaviors that matter in this context are chase rate and whiff rate. Zone swing rate shifts uniformly across the league and does not distinguish outperformers from underperformers. Suggesting that some increase in willingness to swing at strikes is a near-universal response to high leverage that carries little predictive weight on its own.

The unexplained variance in the present analysis deserves direct attention rather than treatment as a residual caveat. The R^2 values of 6.81% in the regression and below 4% in the behavioral ANOVAs indicate that the behavioral metrics evaluated here capture only a small portion of what determines performance under pressure. The remainder is shaped by a large number of factors not present in this study. Possibly the most important is pitcher quality: high leverage plate appearances typically occur against the opposing team's best relief pitchers. Beyond pitcher quality, several other sources of unmeasured variance lives in the game context that aggregate metrics flatten. Pitch usage, count distribution, and the interaction of pitch usage by count are all plausible contributors. Base state and outs introduce a parallel layer that may reshape both pitcher approach and hitter incentives in ways that overlap with leverage but are not identical to it. Contact quality by pitch type and location is similarly absent from the present model, leaving open whether performance changes are concentrated against specific pitches or zones rather than distributed evenly. Combined with individual baseline skill differences and the sample size asymmetry between conditions (298 high-leverage plate appearances per player versus 1,375 low-leverage), these factors offer plausible explanations for why a real but partial signal yields only modest R^2 values.

Several future direction follow from these limitation. The most consequential is modeling performance differentials with pitcher quality as a covariate. This would separate the situational pressure effect from the matchup-difficulty effect. Expanding the response variable to include contact quality by pitch type and location would clarify whether outperformers maintain hard contact alongside plate discipline and potentially whether underperformers make low quality contact on balls or pitches at the edge of the zone. The most important extension, however, is longitudinal. If clutch performance is best understood as behavioral stability, it should persist across seasons rather than year-to-year noise. A multi-season analysis of each player's high-versus-low leverage behavioral drift would test whether stability under pressure is a genuine individual trait. This would test whether these players are able to consistently overperform in the clutch over the course of multiple seasons or if these results are driven by a few lucky hot streaks.

Within the scope of the present study, the actionable recommendation is straightforward. A hitter's clutch performance is best evaluated as deviation from their own baseline rather than absolute outcomes, with chase rate and whiff serving as the most informative behavioral indicators. The hitter who maintains their approach under pressure is the hitter the data identifies as relatively successful in high leverage. A modest but defensible conclusion with direct application to player evaluation, in-game management, and behavioral skill development.

Appendix A
Data Cleaning and Preparation

The following code documents the data preparation pipeline used to construct the analytical dataset from the raw TruMedia Networks export. The pipeline proceeds in five stages: (1) loading the raw season-level split file, (2) renaming columns to a consistent snake-case convention separating overall, high-leverage (`hl_`), and low-leverage (`ll_`) splits, (3) stripping the `%` symbol from rate statistics and coercing them to numeric, (4) computing league averages and the within-player differences used as the response and predictors throughout Sections II–V, and (5) constructing the long-format frame used by the repeated-measures ANOVA in Section III.

Listing 1: Data cleaning and variable construction.

```
---
title: "data_acquisition"
output: html_document
---

```{r}
# Load libraries
library(tidyverse)
library(lubridate)
```

```{r}
# Load data
df <-
    read_csv("C:/STAT_614/project/data/SituationsData2325.cs
```

```{r}
# First look
glimpse(df); head(df); names(df)
```

```{r}
# Standardize column names: overall, HL split,
    LL split
df <- df %>%
  rename(
    # Overall
    player  = `BATTER`, pa      = `PA`,
    ab      = `AB`,     pitches = `PITCHES`,
    z_swing = `Z-SWING`, chase  = `CHASE`,
    whiff   = `WHIFF`,  contact = `CONTACT%`,
    xobp    = `XOBP`,   xslg    = `XSLG`,
    xops    = `XOPS`,
    # High Leverage
    hl_pa   = `HL PA`, hl_ab   = `HL AB`,
    hl_pitches = `HL PITCHES`,hl_z_swing = `HL
        ZSWING`,
    hl_chase = `HL CHASE`, hl_whiff = `HL WHIFF`,
    hl_contact = `HL CONTACT %`,
    hl_xobp = `HL XOBP`, hl_xslg = `HL XSLG`,
    hl_xops = `HL XOPS`,
    # Low Leverage
    ll_pa   = `LL PA`, ll_ab   = `LL AB`,
    ll_pitches = `LL PITCHES`,ll_z_swing = `LL
        Z-SWING`,
    ll_chase = `LL CHASE`, ll_whiff = `LL WHIFF`,
    ll_contact = `LL CONTACT %`,
    ll_xobp = `LL XOBP`, ll_xslg = `LL XSLG`,
    ll_xops = `LL XOPS`
  )
```

```{r}
# Strip "%" from rate columns and coerce to
    numeric
pct_cols <- c(
  "z_swing","chase","whiff","contact",
  "hl_z_swing","hl_chase","hl_whiff","hl_contact",
  "ll_z_swing","ll_chase","ll_whiff","ll_contact"
)

df <- df %>%
  mutate(across(
    all_of(pct_cols),
    ~ as.numeric(str_remove(.x, "%"))
  ))

df %>% select(all_of(pct_cols)) %>% summary()
```

```{r}
# League average table (HL and LL means for
    every metric)
df_league <- df %>%
  summarise(
    hl_pa_avg   = mean(hl_pa, na.rm = TRUE),
    hl_ab_avg   = mean(hl_ab, na.rm = TRUE),
    hl_pitches_avg = mean(hl_pitches, na.rm =
        TRUE),
    hl_z_swing_avg = mean(hl_z_swing, na.rm =
        TRUE),
    hl_chase_avg = mean(hl_chase, na.rm = TRUE),
    hl_whiff_avg = mean(hl_whiff, na.rm = TRUE),
    hl_contact_avg = mean(hl_contact, na.rm =
        TRUE),
    hl_xobp_avg = mean(hl_xobp, na.rm = TRUE),
```
```

```
    hl_xslg_avg = mean(hl_xslg, na.rm = TRUE),
    hl_xops_avg = mean(hl_xops, na.rm = TRUE),
    ll_pa_avg  = mean(ll_pa, na.rm = TRUE),
    ll_ab_avg  = mean(ll_ab, na.rm = TRUE),
    ll_pitches_avg = mean(ll_pitches, na.rm = TRUE),
    ll_z_swing_avg = mean(ll_z_swing, na.rm = TRUE),
    ll_chase_avg = mean(ll_chase, na.rm = TRUE),
    ll_whiff_avg = mean(ll_whiff, na.rm = TRUE),
    ll_contact_avg = mean(ll_contact, na.rm = TRUE),
    ll_xobp_avg = mean(ll_xobp, na.rm = TRUE),
    ll_xslg_avg = mean(ll_xslg, na.rm = TRUE),
    ll_xops_avg = mean(ll_xops, na.rm = TRUE)
  )

# League difference table (HL - LL, reference only)
df_league_diff <- tibble(
  metric = c("PA","AB","Pitches","Z-Swing%","Chase%",
             "Whiff%","Contact%","xOBP","xSLG","xOPS"),
  hl_avg = c(df_league$hl_pa_avg,
             df_league$hl_ab_avg,
             df_league$hl_pitches_avg,
             df_league$hl_z_swing_avg,
             df_league$hl_chase_avg,
             df_league$hl_whiff_avg,
             df_league$hl_contact_avg,
             df_league$hl_xobp_avg,
             df_league$hl_xslg_avg,
             df_league$hl_xops_avg),
  ll_avg = c(df_league$ll_pa_avg,
             df_league$ll_ab_avg,
             df_league$ll_pitches_avg,
             df_league$ll_z_swing_avg,
             df_league$ll_chase_avg,
             df_league$ll_whiff_avg,
             df_league$ll_contact_avg,
             df_league$ll_xobp_avg,
             df_league$ll_xslg_avg,
             df_league$ll_xops_avg)
) %>%
  mutate(diff = hl_avg - ll_avg)

print(df_league_diff)
```

```{r}
# Wide table: derived response and predictor variables

# League mean declines (used to centre the adjusted differential)
league_mean_diff_xobp <- df_league$hl_xobp_avg - df_league$ll_xobp_avg
league_mean_diff_xslg <- df_league$hl_xslg_avg - df_league$ll_xslg_avg
league_mean_diff_xops <- df_league$hl_xops_avg - df_league$ll_xops_avg

df_wide <- df %>%
  mutate(
    # Performance differentials
    clutch_diff_xobp = hl_xobp - ll_xobp,
    clutch_diff_xslg = hl_xslg - ll_xslg,
    clutch_diff_xops = hl_xops - ll_xops,

    # Adjusted differentials (relative to league mean)
    clutch_diff_xobp_adj = clutch_diff_xobp - league_mean_diff_xobp,
    clutch_diff_xslg_adj = clutch_diff_xslg - league_mean_diff_xslg,
    clutch_diff_xops_adj = clutch_diff_xops - league_mean_diff_xops,

    # Behavioural differentials (percentage points)
    delta_z_swing = hl_z_swing - ll_z_swing,
    delta_chase = hl_chase - ll_chase,
    delta_whiff = hl_whiff - ll_whiff,
    delta_contact = hl_contact - ll_contact,

    # Median split on adjusted xOPS differential
    clutch_group = factor(ifelse(
      clutch_diff_xops_adj >=
        median(clutch_diff_xops_adj, na.rm = TRUE),
      "Above Average", "Below Average"
    ))
  )

df_wide %>%
  select(player, clutch_diff_xops, clutch_diff_xops_adj,
         delta_chase, clutch_group) %>%
  head(10)

table(df_wide$clutch_group)
summary(df_wide$clutch_diff_xops_adj)
```

```{r}
# Long table for repeated-measures ANOVA
df_long <- df_wide %>%
  select(player, clutch_group,
         hl_pa, hl_ab, hl_pitches, hl_z_swing, hl_chase,
         hl_whiff, hl_contact, hl_xobp, hl_xslg, hl_xops,
         ll_pa, ll_ab, ll_pitches, ll_z_swing, ll_chase,
         ll_whiff, ll_contact, ll_xobp, ll_xslg, ll_xops) %>%
  pivot_longer(
    cols      = -c(player, clutch_group),
    names_to  = c("leverage", ".value"),
    names_pattern = "(hl|ll)_(.*)"
  ) %>%
  mutate(leverage = factor(
    case_match(leverage, "hl" ~ "High", "ll" ~ "Low"),
    levels = c("Low", "High")
  ))

glimpse(df_long); head(df_long, 6)
nrow(df_long) == 2 * n_distinct(df_wide$player)
```

```{r}
# Export cleaned tables
output_path <- "C:/STAT_614/project/data/"
write_csv(df_wide, paste0(output_path, "df_wide.csv"))
write_csv(df_long, paste0(output_path, "df_long.csv"))
write_csv(df_league, paste0(output_path, "df_league.csv"))
write_csv(df_league_diff, paste0(output_path, "df_league_diff.csv"))
cat("All tables saved to", output_path, "\n")
```

Appendix B
Supplementary Statistical Output

This appendix reproduces the Minitab output that supports the assumption checks, sensitivity analyses, and supplementary statistics referenced throughout Sections II–V. Each subsection corresponds to a single test or check in the main text. Where the underlying tables are too wide to typeset in a single column, they have been reformatted as `booktabs` tables; the numerical content is unchanged.

A. Section II: Paired t-Test on xOPS

The paired *t*-test compared each player's high-leverage and low-leverage xOPS, treating each hitter as their own control. The descriptive summary in Table III shows the unconditional means used to compute the paired difference, and Table IV reports the paired estimate, 95% confidence interval, and two-sided *t*-statistic.

TABLE III: Section II descriptive statistics for paired *t*-test.

Sample	N	Mean	StDev	SE Mean
hl_xops	143	0.72508	0.08759	0.00732
ll_xops	143	0.75761	0.07736	0.00647

TABLE IV: Section II paired-difference estimate and test ($\mu_d = \mu_{HL} - \mu_{LL}$).

Mean$_d$	StDev$_d$	SE Mean	95% CI	T	p
-0.0325	0.0623	0.0052	$(-0.0428, -0.0222)$	-6.24	< 0.001

a) Normality of within-player differences: The Anderson–Darling test on the within-player xOPS differences returned a p-value of 0.327, providing no evidence at the 5% level that the differences depart from normality. Combined with the approximately linear normal probability plot (Fig. 1), this supports treating the paired differences as approximately normal.

b) Effect size: Cohen's d for paired samples, computed as the mean within-player difference divided by the standard deviation of those differences, yields

$$d = \frac{-0.0325}{0.0623} \approx -0.522,$$

which matches the value reported in Section II-D and corresponds to a moderate standardized effect.

c) Outlier sensitivity: The box plot in Fig. 2 identified two outliers (one high, one low). The paired *t*-test was re-run with these two observations removed ($n = 141$). Tables V and VI report the resulting descriptive statistics and test output. The mean difference, 95% confidence interval, and *t*-statistic are essentially unchanged, confirming that the Section II conclusion is not driven by the outlying differences.

B. Section III: Repeated-Measures ANOVA

A separate repeated-measures ANOVA was conducted for Zone Swing Rate, Chase Rate, and Whiff Rate. With only two levels of leverage, sphericity is automatically satisfied and pooled-variance ANOVA was used in each case. Table VII consolidates the three Analysis-of-Variance tables, and Table VIII reports the marginal means used to compute the directional shifts.

TABLE V: Section II outlier sensitivity — descriptive statistics ($n = 141$).

Sample	N	Mean	StDev	SE Mean
hl_xops	141	0.72469	0.08814	0.00742
ll_xops	141	0.75691	0.07439	0.00626

TABLE VI: Section II outlier sensitivity — paired-difference test ($n = 141$).

Mean$_d$	StDev$_d$	SE Mean	95% CI	T	p
-0.0322	0.0581	0.0049	$(-0.0419, -0.0226)$	-6.59	< 0.001

TABLE VII: Section III analysis of variance, by metric.

Source	DF	Adj SS	Adj MS	F	p
Zone Swing Rate					
leverage	1	352.0	352.03	10.83	0.001
Error	284	9,231.9	32.51		
Total	285	9,583.9			
Chase Rate					
leverage	1	218.5	218.53	5.96	0.015
Error	284	10,420.9	36.69		
Total	285	10,639.4			
Whiff Rate					
leverage	1	191.5	191.45	6.08	0.014
Error	284	8,943.6	31.49		
Total	285	9,135.1			

TABLE VIII: Section III marginal means by leverage condition (%).

Metric	Leverage	N	Mean	StDev	95% CI
Zone Swing	High	143	68.578	5.793	$(67.639, 69.516)$
Zone Swing	Low	143	66.359	5.608	$(65.420, 67.297)$
Chase	High	143	30.555	6.494	$(29.557, 31.552)$
Chase	Low	143	28.806	5.587	$(27.809, 29.803)$
Whiff	High	143	22.997	5.773	$(22.073, 23.921)$
Whiff	Low	143	21.361	5.446	$(20.437, 22.285)$

The model-summary statistics (pooled S, R^2, adjusted R^2, predicted R^2) are reproduced in Table IX. The small R^2 values reinforce the discussion in Section III: leverage produces a statistically detectable shift in approach, but explains only a small fraction of the cross-player variation in plate-discipline behaviour.

a) Normality of within-player differences: Anderson–Darling tests on the within-player differences for each of the three behavioural metrics returned p-values above 0.05 in every case, consistent with the approximately linear normal probability plots in Fig. 3. The within-player differences can therefore be treated as approximately normal for inference.

b) Outlier sensitivity: The box plots in Fig. 4 identified two high outliers in the Chase Rate differences and one in the Whiff Rate differences. Re-running each ANOVA with the corresponding outliers removed left the directional conclusions

TABLE IX: Section III model summary statistics.

Metric	S	R^2	R^2_{adj}	R^2_{pred}
Zone Swing	5.70145	3.67%	3.33%	2.31%
Chase	6.05750	2.05%	1.71%	0.67%
Whiff	5.61174	2.10%	1.75%	0.71%

and the rejection of H_0 unchanged: Zone Swing remained significant at the 0.001 level, and Chase and Whiff Rate remained significant at the 0.05 level. The reported F-statistics are not artefacts of the extreme observations.

C. Section IV: Multiple Linear Regression

The regression in Section IV models the league-adjusted xOPS differential ($\Delta\text{xOPS}_{\text{adj}}$) as a linear function of the three behavioural change variables $\Delta\text{ZoneSwing}$, ΔChase, and ΔWhiff. Tables X and XI reproduce the coefficient and ANOVA tables from Minitab, and Table XII reports the model summary.

TABLE X: Section IV regression coefficients.

Term	Coef	SE Coef	T	p	VIF
Constant	0.01011	0.00777	1.30	0.195	–
ΔZ-Swing	0.00214	0.00197	1.09	0.279	1.08
ΔChase	−0.00376	0.00202	−1.86	0.065	1.11
ΔWhiff	−0.00507	0.00236	−2.15	0.033	1.03

The fitted equation is

$$\begin{aligned}\widehat{\Delta\text{xOPS}}_{\text{adj}} &= 0.01011 + 0.00214\,\Delta\text{ZoneSwing} \\ &\quad - 0.00376\,\Delta\text{Chase} - 0.00507\,\Delta\text{Whiff}.\end{aligned}$$

TABLE XI: Section IV analysis of variance.

Source	DF	Adj SS	Adj MS	F	p
Regression	3	0.037550	0.012517	3.38	0.020
ΔZ-Swing	1	0.004359	0.004359	1.18	0.279
ΔChase	1	0.012820	0.012820	3.47	0.065
ΔWhiff	1	0.017056	0.017056	4.61	0.033
Error	139	0.513986	0.003698		
Total	142	0.551536			

TABLE XII: Section IV model summary.

S	R^2	R^2_{adj}	R^2_{pred}
0.0608090	6.81%	4.80%	1.28%

a) Multicollinearity: All three predictor VIF values lie between 1.03 and 1.11 (Table X), well below the conventional threshold of 5–10. The plate-discipline change variables therefore carry largely independent information, so the small overall R^2 is not the result of redundant predictors competing for the same variance.

b) Residual diagnostics: The Anderson–Darling test on the regression residuals returned a p-value of 0.784, providing no evidence at the 5% level against normality. Combined with the residuals-vs-fitted plot (Fig. 5a) — which shows residuals scattered around zero with no funnel or curvature — and the normal probability plot (Fig. 5b), the assumptions of linearity, constant variance, and approximately normal residuals are satisfied.

D. Section V: Equality-of-Variance Tests

Before applying two-sample *t*-tests, the equality-of-variance assumption was tested for each behavioural metric using both the Bonett and Levene methods. Tables XIII and XIV summarise the results. The variance test for ΔZone Swing returned a p-value below 0.05 under both methods (Bonett 0.044, Levene 0.036), indicating unequal variances; Welch's *t*-test was therefore applied to that metric. The variance tests for ΔChase (Bonett $p = 0.501$, Levene $p = 0.593$) and ΔWhiff (Bonett $p = 0.771$, Levene $p = 0.715$) produced p-values comfortably above 0.05 under both methods, supporting the equal-variance assumption and the use of the pooled-variance two-sample *t*-test for those metrics.

TABLE XIII: Section V within-group standard deviations by metric.

Metric	Group	N	StDev	Var.	95% CI for σ
ΔZ-Swing	Above Avg.	72	2.948	8.693	(2.626, 3.403)
ΔZ-Swing	Below Avg.	71	2.401	5.765	(2.070, 2.865)
ΔChase	Above Avg.	72	2.752	7.575	(2.190, 3.555)
ΔChase	Below Avg.	71	2.486	6.181	(2.149, 2.958)
ΔWhiff	Above Avg.	72	2.127	4.523	(1.825, 2.548)
ΔWhiff	Below Avg.	71	2.208	4.875	(1.839, 2.726)

TABLE XIV: Section V variance-equality tests (H_0: $\sigma_1/\sigma_2 = 1$).

Metric	Method	Stat.	DF1	DF2	p
ΔZ-Swing	Bonett	—	—	—	0.044
ΔZ-Swing	Levene	4.51	1	141	0.036
ΔChase	Bonett	—	—	—	0.501
ΔChase	Levene	0.29	1	141	0.593
ΔWhiff	Bonett	—	—	—	0.771
ΔWhiff	Levene	0.13	1	141	0.715

a) Within-group normality: Anderson–Darling tests for each behavioural change variable, computed separately within the Above-Average and Below-Average performance groups, all returned p-values above 0.05 (six tests in total). The within-group distributions can therefore be treated as approximately normal, satisfying the second-sample-mean assumption for all three two-sample *t*-tests.

E. Section V: Two-Sample t-Tests by Performance Group

The two-sample *t*-tests compared each behavioural change variable between Above-Average and Below-Average performers, where group membership was assigned by a median split on $\Delta\text{xOPS}_{\text{adj}}$ (see Listing 1). Welch's test was applied to ΔZone Swing (unequal variances), and the pooled-variance test was applied to ΔChase and ΔWhiff (equal variances supported by the Bonett and Levene tests above).

The negative differences for ΔChase and ΔWhiff indicate that Above-Average performers exhibited smaller increases in chase and whiff rates than Below-Average performers under high leverage; the ΔZone Swing difference is small

TABLE XV: Section V two-sample descriptive statistics.

Metric	Group	N	Mean	StDev	SE
ΔZ-Swing	Above Avg.	72	2.06	2.95	0.35
ΔZ-Swing	Below Avg.	71	2.38	2.40	0.28
ΔChase	Above Avg.	72	1.24	2.75	0.32
ΔChase	Below Avg.	71	2.26	2.49	0.30
ΔWhiff	Above Avg.	72	1.23	2.13	0.25
ΔWhiff	Below Avg.	71	2.05	2.21	0.26

TABLE XVI: Section V two-sample *t*-test results (H_0: $\mu_{\text{Above}} - \mu_{\text{Below}} = 0$).

Metric	Diff.	95% CI	T	DF	p
ΔZ-Swing (Welch)	-0.318	$(-1.206,\ 0.571)$	-0.71	136	0.481
ΔChase (Pooled)	-1.020	$(-1.887,\ -0.153)$	-2.33	141	0.021
ΔWhiff (Pooled)	-0.829	$(-1.545,\ -0.112)$	-2.29	141	0.024

in magnitude and statistically indistinguishable from zero. Together these results provide the behavioural separation between performance groups discussed in Section V-D.

REFERENCES

[1] MLB Advanced Media, “Leverage index (LI),” MLB.com Glossary, 2025, accessed: April 2025. [Online]. Available: https://www.mlb.com/glossary/advanced-stats/leverage-index

[2] TruMedia Networks, “Trumedia networks baseball analytics platform,” TruMedia Networks, 2025, accessed: April 2025. [Online]. Available: https://www.trumedianetworks.com/baseball